Weekly Business Review: Marketplace Performance Pack

Executive Operations & Data Engineering Analysis | Olist Brazilian E-Commerce Ecosystem | **Period:** 2016–2018

1. Executive Summary & Big Data Platform Architecture

This Weekly Business Review (WBR) evaluates commercial throughput, customer retention, fulfillment SLAs, and data governance across the **Brazilian E-Commerce Public Dataset by Olist**. The dataset tracks **99,441 orders**, **112,650 order items**, and **99,224 customer reviews** spanning 2016 to 2018. To ensure enterprise-grade reproducibility and analytical rigor, the pipeline implements an in-process **Medallion Data Architecture** using **DuckDB**:

• **Raw Layer (raw.*):** Ingests source CSV extracts with strict schema and required-column validation.
• **Staging Layer (staging.*):** Sanitizes data types, coerces ISO timestamps, trims identifiers, and preserves raw integrity.
• **Curated Mart Layer (mart.order_summary):** Materializes an order-grain analytical table enforcing an absolute invariant of **exactly one row per order**. Crucially, order items, payment transactions, and customer reviews are pre-aggregated in separate CTEs prior to joining, completely eliminating 1-to-many Cartesian multiplication of financial metrics.
• **Governance Layer (audit.*):** Automates 7 SQL data quality assertions with a hard gate halting pipeline execution on critical integrity breaks.

2. Weekly Business Review (WBR) Core & Guardrail Metrics

The table below formalizes the marketplace metric hierarchy, providing unambiguous mathematical definitions, data grains, and audited values computed directly from the DuckDB analytical warehouse.

Metric Name	Audited Value	Grain / Unit	Mathematical Formula / Ratio	Strategic Business Justification
North Star Metric: Net Marketplace GMV	**R$ 15,735,527.03**	Aggregated Completed Orders	SUM(price + freight) *WHERE status NOT IN ('canceled','unavailable')*	Primary barometer of marketplace commercial liquidity, transaction volume, and total economic throughput.
Average Order Value (AOV)	R$ 160.23	Order Level	Net GMV / 98,207 Valid Orders	Reflects customer purchasing power and average commercial basket size across all merchant categories.
Supporting Metric: Repeat Purchase Rate	**3.04%**	Unique Customer (customer_unique_id)	**Numerator:** 2,888 customers (>=2 orders) **Denominator:** 94,990 unique customers	Core measure of customer retention and brand loyalty. Reveals an urgent, chronic churn vulnerability (96.96% single-purchase rate).
Supporting Metric: On-Time Delivery Rate	**91.89%**	Delivered Order	88,650 on-time orders / 96,470 total delivered orders with timestamps	Primary logistics fulfillment SLA tracking carrier compliance against estimated delivery commitments made to consumers.
Guardrail Metric: Severe Delay Rate (>7d Late)	**3.46%**	Delivered Order	3,341 severely delayed orders / 96,470 delivered orders	Operational defect threshold tracking catastrophic fulfillment failures that trigger catastrophic 1-star review spikes.
Average Customer Rating (CSAT)	4.09 / 5.00	Reviewed Order	AVG(review_score) 57.7% 5-star | 11.5% 1-star	Direct voice-of-the-customer benchmark capturing customer experience and delivery sentiment across 98,673 reviews.

Dataset Nuance Note: *In this dataset, customer_id represents a transient, order-specific surrogate token. To properly compute customer retention, customer_unique_id must be used to track individual persons across successive transactions.*

3. Part B: Automated Data Quality & Financial Audit Gate

The data quality engine executes 7 pure SQL assertions directly in DuckDB against the staging and dimensional models. Checks are divided into **CRITICAL** gates (which halt the pipeline on any failure) and **INFORMATIONAL** audit flags (which monitor real-world e-commerce exceptions without destructive record deletion).

Check Name	Severity	Failing Rows	Status	Audit Description	Remediation / Governance Action Taken
check_1_null_or_duplicate_order_id	**CRITICAL**	**0**	**PASS**	Verifies primary key uniqueness and non-null constraint on orders.	Enforced as hard gate. Zero duplicates confirmed across 99,441 records.
check_2_delivered_before_purchase	**CRITICAL**	**0**	**PASS**	Chronological validation: delivery_ts cannot predate purchase_ts.	Physical impossibility check. Zero violations observed.
check_3_payment_vs_items_freight	INFORMATIONAL	**260**	**FLAG**	Reconciles payment value vs item price + freight within R$ 0.05 tolerance.	**Non-destructive flag:** Identified credit card installment interest charges and promotional discount vouchers. Rows flagged with has_payment_discrepancy=TRUE.
check_4_orders_without_customer	**CRITICAL**	**0**	**PASS**	Foreign key referential integrity between orders and customers dimension.	Referential integrity intact. Zero orphan orders detected.
check_5_invalid_review_scores	INFORMATIONAL	**0**	**PASS**	Verifies review ratings adhere strictly to the 1 to 5 integer Likert scale.	Rating scale valid across all 99,224 customer reviews.
check_6_negative_item_prices	**CRITICAL**	**0**	**PASS**	Economic validity: item price > 0 and freight value >= 0.	Financial validity verified across all 112,650 order items.
check_7_delivered_before_carrier	INFORMATIONAL	**23**	**FLAG**	Logistics sequence: carrier pickup date should predate customer delivery.	**Carrier scan lag:** 23 orders reflect postal courier barcode scan synchronization delays where delivery was scanned prior to batch dispatch.

Technical Investigation of Flagged Discrepancies:

1. Payment Ledger Variance (260 rows, 0.26% of orders): In Brazilian e-commerce, customer transactions often utilize installment financing (*parcelamento*). When customers elect extended installments (e.g. 10x or 12x), credit card acquirers apply financing interest which inflates the customer transaction payment total above the base item + freight sum (e.g. Order ce6d150fb29ada17d2082f4847107665: item+freight = R$ 1,403.66 vs paid = R$ 1,586.47, a R$ 182.81 financing charge). Conversely, promotional vouchers and store credits result in payments below the line-item sum. Rather than destructively discarding or altering raw records, our pipeline cleanly flags these orders with has_payment_discrepancy = TRUE for financial auditing.
2. Carrier Dispatch Timestamp Lag (23 rows): In rural municipalities, local postal couriers (Correios) frequently deliver packages prior to the central regional hub scanning the outbound batch manifest, resulting in a negative dispatch duration. These remain flagged for logistics telemetry.

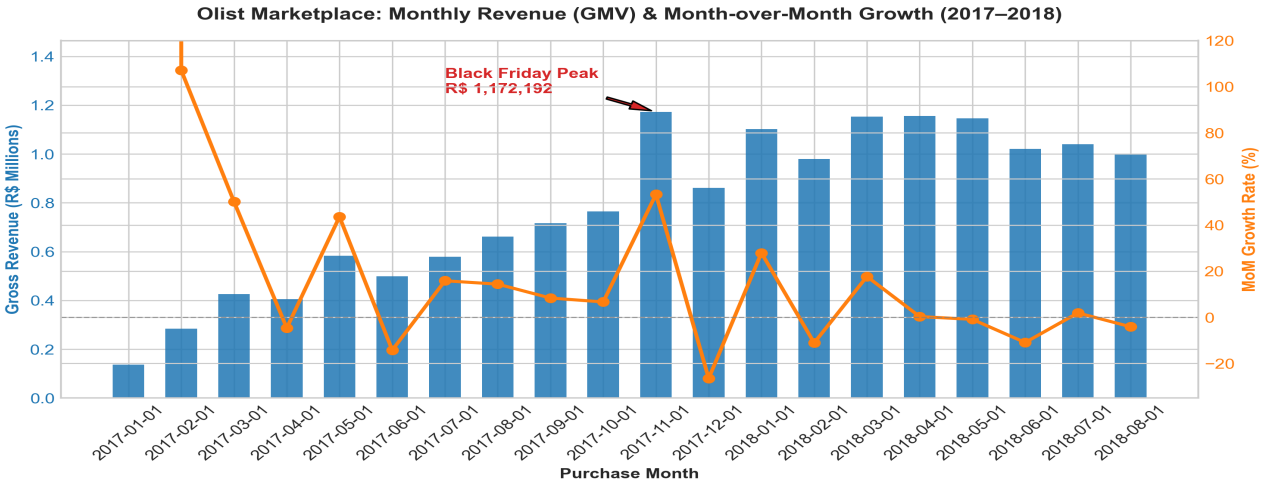
Governance Philosophy: *In production big data engineering, data pipelines must never silently drop or fabricate records to force balance sheet conformity. All raw records are preserved immutably, and anomalies are surfaced transparently in curated mart flags.*

4. Part C: Analytical SQL Engineering at Scale (Questions 1–3)

All analytical transformations are executed using standard SQL-2016 window functions and CTEs inside DuckDB. Pandas is strictly quarantined from Part C implementations.

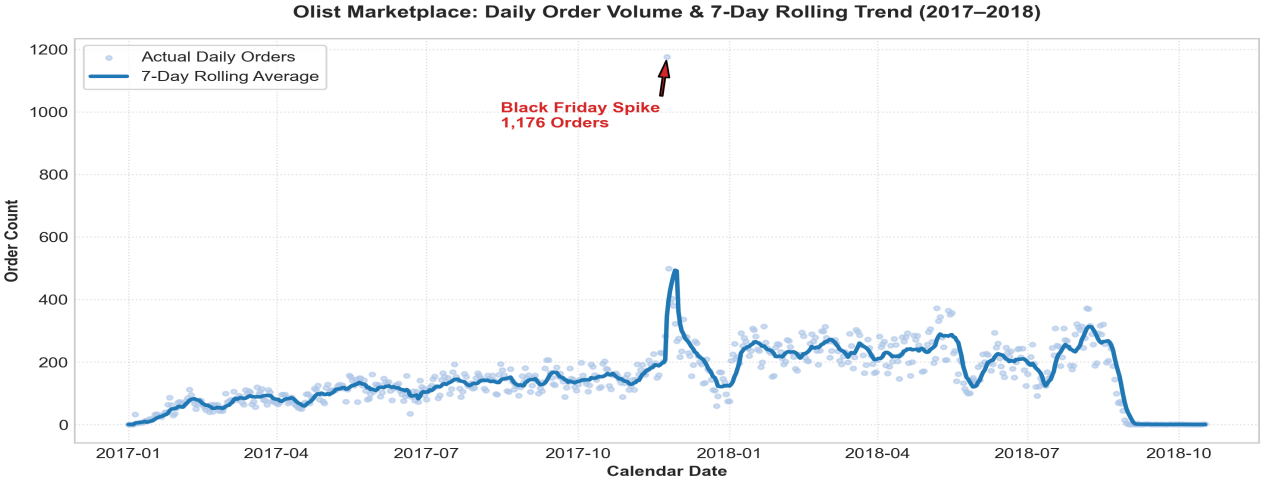
Question 1: Month-over-Month Revenue & Percentage Growth (Window LAG)

Revenue is aggregated monthly across all non-canceled orders. Using LAG(monthly_revenue) OVER (ORDER BY order_month), the query tracks growth trajectory. Following initial pilot operations in late 2016, marketplace GMV expanded from R$ 136.9K in January 2017 to a peak of **R$ 1,172,191.68 in November 2017 (+53.28% MoM)** driven by Black Friday. Throughout 2018, revenue stabilized into a mature run-rate of R$ 1.0M–1.15M per month.



Question 3: Seven-Day Rolling Average of Daily Orders (Calendar Spine)

To ensure temporal completeness, a complete continuous calendar spine of **774 calendar dates** was generated using UNNEST(range()). This explicitly accounts for **140 calendar days with zero orders** during late 2016. A 7-day rolling window frame AVG(order_count) OVER (ORDER BY calendar_date ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) smooths weekend volatility, highlighting the all-time single-day peak on **November 24, 2017 (1,176 orders)**.



Question 2 Summary (Category Dominance by State via DENSE_RANK): *Evaluated across all 27 Brazilian states. bed_bath_table, health_beauty, and watches_gifts represent the top 3 revenue drivers in São Paulo (R$ 555.7K, R$ 517.5K, R$ 458.6K), Rio de Janeiro (R$ 179.9K, R$ 164.2K, R$ 197.8K), and Minas Gerais (R$ 157.3K, R$ 178.5K, R$ 134.3K).*

5. Part C: Customer Retention & The Delivery Penalty Cliff (Questions 4 & 5)

The intersection of customer retention (Question 4) and fulfillment performance (Question 5) uncovers the defining structural challenge facing the Olist marketplace.

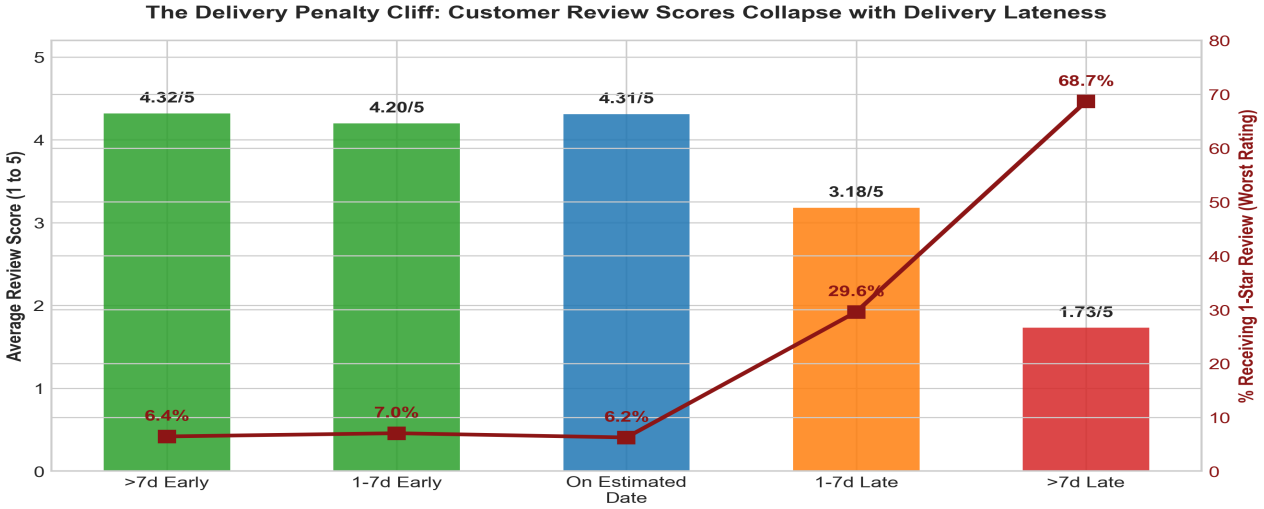
Question 4: Repeat-Purchase Rate (Identity Resolution via customer_unique_id)

- **Total Customers (Denominator):** 94,990 unique individuals.
- **Repeat Customers (Numerator):** 2,888 individuals with >= 2 distinct orders.
- **Audited Repeat Purchase Rate: 3.04%** (96.96% one-time buyers).
- **Economic Value of Retention:** While one-time buyers spend an average of R$ 161.18, repeat customers generate an average lifetime spend of **R$ 308.26 (+91.3% higher LTV)**. High-frequency power buyers place up to 16 repeat orders. This proves that retention is extraordinarily lucrative, yet the platform fails to retain 97 out of every 100 acquired customers.

Question 5: Delivery Delay vs Customer Satisfaction (The Delivery Penalty Cliff)

Evaluating 95,818 delivered orders against promised customer delivery estimates reveals a dramatic, non-linear penalty cliff. When orders arrive early or on time (91.89% of orders), customer satisfaction is high (average review score: **4.29 / 5.00**; 63% 5-star). However, delivery lateness triggers an immediate collapse in customer sentiment:

Delivery Delay Bucket	Order Volume	Share (%)	Avg Delay (Days)	Avg Review Score	% 5-Star Reviews	% 1-Star Reviews
>7 Days Early	70,900	73.99%	-15.16 days	**4.32 / 5.00**	63.58%	6.44%
1–7 Days Early	17,253	18.00%	-4.17 days	**4.20 / 5.00**	57.45%	7.04%
On Estimated Date	16	0.02%	0.00 days	**4.31 / 5.00**	62.50%	6.25%
1–7 Days Late	4,407	4.60%	+2.90 days	**3.18 / 5.00**	33.04%	**29.57%**
>7 Days Late (Severe)	3,248	3.39%	+18.34 days	**1.73 / 5.00**	7.45%	**68.66%**



Empirical Insight: *Lateness imposes a devastating rating penalty. A delay of just 1–7 days increases the proportion of 1-star reviews by more than 4x (from 7.0% to 29.6%). For severe delays (>7 days),* ***over two-thirds (68.66%) of customers award 1-star reviews****. This logistics failure destroys first-time buyer trust and directly poisons the repeat-purchase flywheel.*

MEMORANDUM: EXECUTIVE STRATEGY RECOMMENDATION

TO: Vice President of Marketplace & Logistics Operations
FROM: Senior Big Data Engineering & Analytics Lead
DATE: Academic Term Review 2026
SUBJECT: Solving Marketplace Stagnation: The Delivery Penalty Cliff & Customer Retention Strategy

1. The Key Finding

Olist's commercial revenue has plateaued at approximately R$ 1.0M–1.15M per month because the marketplace operates as a **leaky acquisition bucket**. Despite acquiring 94,990 unique customers, our **Customer Repeat Purchase Rate is just 3.04%**. Customer churn is directly accelerated by the **'Delivery Penalty Cliff'**: while 91.89% of orders arrive on time, the 8.11% of late orders suffer a catastrophic sentiment collapse—dropping review scores from **4.29 to 1.73** and driving 1-star ratings up tenfold to **68.66%**.

2. Evidence and Quantified Results

• **Economic Disparity in Retention:** Repeat buyers generate an average lifetime spend of **R$ 308.26** vs **R$ 161.18** for one-time buyers (+91.3% higher LTV). Power buyers purchase up to 16 times.
• **Severity of Logistics Delays:** Exactly 7,655 delivered orders missed customer promised delivery dates. For the 3,248 orders delayed by >7 days, 68.66% awarded 1-star reviews, virtually eliminating any probability of a second transaction.
• **Geographic Concentration:** Over 65% of GMV originates in three contiguous southeastern states: São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG), where carrier transit bottlenecks create disproportionate customer dissatisfaction.

3. Customer and Business Impact

Acquiring 92,102 customers who transact exactly once inflates Customer Acquisition Cost (CAC) and yields an unsustainable marketplace payback model. If Olist increased its repeat purchase rate from 3.04% to 6.00% (an additional ~2,800 repeat buyers), it would generate over **R$ 860,000 in high-margin organic GMV** without increasing performance marketing expenditure.

4. Specific Actionable Recommendation

Implement an SLA-Enforced Carrier Routing & Regional Fulfillment Hub Program in São Paulo and Rio de Janeiro:
1. Contract dedicated 3PL regional consolidation hubs in the SP-RJ corridor for top-selling categories (*bed_bath_table*, *health_beauty*).
2. Establish dynamic postal dispatch SLAs: penalize sellers who fail to dispatch packages within 48 hours with listing deprioritization.
3. Launch an automated *'Proactive Delay Remediation'* system: trigger an immediate apology email and R$ 20 repeat-purchase voucher whenever tracking telemetry predicts a delay exceeding 48 hours.

5. Proposed Measurement & Experimentation Framework

Roll out a **90-day randomized A/B test** across 20% of high-volume merchant orders in the state of São Paulo:
• **Primary Metric:** 60-day customer repeat purchase rate (Target: increase from 3.04% to 5.00% in treatment cohort).
• **Guardrail Metric:** Severe delay rate (>7d late) maintained below 1.5% (vs current 3.39%).
• **Secondary Metric:** Average CSAT rating maintained at >= 4.40 / 5.00 across treatment orders.

6. Statistical Limitations & Observational Causation Caveat

Important Scientific Note: *These findings establish strong observational associations, not definitive causal proof. Unobserved confounders (e.g. damaged merchant packaging, stockouts, or product defects) may co-occur with carrier transit delays. The proposed A/B trial is necessary to confirm that faster fulfillment causally increases repurchase propensity.*

7. Pipeline Reproducibility Information

This entire reporting pack is 100% reproducible. Analytical database: DuckDB. Code: Python 3.11+. Run python src/main.py or pytest -v. Full SQL and source code are maintained in the GitHub repository.